Thornquist Outfitters Data Lake Technical Report

CCGC 5506 Mid Term Project | Humber Fall 2026 | 22 October 2026

Business purpose and delivery outcome

Thornquist Outfitters needs a shared view of store and ecommerce sales linked to customer information. Its source formats and identifiers differ, and deliberately inconsistent records make direct aggregation unreliable. We built a zoned Amazon S3 lake with three ingestion mechanisms, a Glue validation and conformance job, partitioned Parquet analytics, and an offline notebook visualization layer. The verified reporting period is August 24 to 30, 2026. The sample represents ten stores, although the fictional business operates forty.

The accepted dataset contains 1,601 sales lines and total sales of 320,570.35 in source monetary units. POS contributes 214,561.78 and ecommerce 106,008.57. These values exclude quarantined records. No currency code was supplied, so the analysis does not label amounts as Canadian dollars. The deployment demonstrates the processing stages and several automation components; a complete automatic refresh chain remains unfinished.

Architecture and zone boundaries

The deployment uses bucket thornquist-data-lake-2026102 in us-east-1. Landing retains submitted synthetic source files. Raw holds ingested data before business validation. Trusted contains records that pass validation and the flattened sales model. Curated exposes accepted sales in a query-oriented structure. Separate quarantine and reports prefixes retain rejected records and reconciliation evidence. Query outputs use athena-results. The companion architecture diagram shows data flow separately from catalog, monitoring and access controls.

Promotion is explicit. Ingestion moves Landing data into Raw without treating a successful copy as proof of business validity. The Glue sales job validates Raw inputs and sends invalid records to Quarantine with rejection reasons. Accepted records enter Trusted, where customer enrichment and order reconciliation produce a common sales-line model. Athena CREATE TABLE AS SELECT, or CTAS, creates Curated Parquet from that model. Curated refresh is manually initiated. These logical prefixes organize processing; the shared execution identity means they are not yet independently enforced access boundaries.

Ingestion and catalog design

POS ingestion uses a Bash copy from landing/pos to raw/pos. Ecommerce ingestion uses the thornquist-ingest-ecommerce Lambda function, triggered by S3 object creation under landing/ecommerce. Manual invocation and an actual upload-triggered copy were verified. The original JSON array contains 265 orders and remains intact in raw/ecommerce. No API Gateway endpoint was deployed.

The thornquist-customer-ingestion Glue job reads the customer CSV and writes 400 rows as Parquet under raw/customers. This makes the customer Raw representation structurally transformed, while Landing preserves its original file. The daily job trigger is active at 07:00 UTC. A separate manual conversion produces ecommerce NDJSON for Athena catalog exploration; the sales ETL reads the original JSON array, so it does not depend on that conversion.

Four crawlers populate the thornquist_outfitters database. Daily schedules were configured for customers at 07:15, ecommerce NDJSON at 07:20, POS at 07:25 and Trusted sales at 07:30 UTC. Existing successful runs

demonstrate crawler operation; successful automatically scheduled runs have not yet been observed. The timers are independent and do not wait for upstream success. Catalog coverage has been demonstrated for Raw sources, Trusted sales and the manually created Curated table, rather than every auxiliary prefix.

Schema reconciliation

We resolved source differences in the common sales model and documented the remaining classification assumptions. First, POS dates accept ISO and day/month/year formats, while ecommerce supplies ISO timestamps. POS dates are normalized, and timezone-aware ecommerce timestamps are converted to UTC. Curated exposes a DATE for reporting. Date-only store records do not acquire a fabricated time of day.

Second, POS cust_num and ecommerce cust_ref become customer_id. Identifiers are treated as strings so leading zeros remain meaningful. The POS crawler initially inferred numeric identifiers; transaction_id, store_id and cust_num were manually corrected to string in the catalog. The crawler was subsequently configured to ignore schema updates, protecting that manual definition. A further crawl followed by an identifier-type regression check remains outstanding.

Third, ecommerce orders contain nested item arrays, whereas each POS record already describes a sales line. Accepted orders are expanded into item lines with order identifiers and line numbers. Transaction counts therefore combine 1,048 POS transactions with 226 ecommerce orders, rather than counting all 1,601 lines as separate transactions.

Fourth, amounts arrive with different numeric representations. Decimal arithmetic and cent rounding support item-total reconciliation; Curated casts unit price and line amount to DECIMAL(18,2). Fifth, POS region describes a store location, while ecommerce shipping_region describes a delivery destination. The common reporting region preserves that interpretation. Sixth, ecommerce categories are absent. For category analysis, trimmed lowercase product names map to POS categories only when the name has exactly one nonblank category. All 553 ecommerce lines matched. This derived query classification leaves the underlying Curated category field unchanged and requires review on refresh.

Automated quality gate and recovery

Custom Python validation inside thornquist-sales-etl implements the permitted alternative to managed Glue Data Quality rulesets. The rules cover required fields and placeholder values, duplicate identifiers, valid dates and timestamps, customer reference integrity, positive quantities, nonnegative finite prices, nonempty item arrays and required item fields. A further check compares each accepted ecommerce order total with the calculated item total. A record can fail more than one rule.

Rejected records retain their original content, source file, record number and rejection reasons. They do not enter accepted sales analytics. The POS walk-in identifier is an explicit guest exception to the customer lookup. It does not represent an individually identified customer. Unrecognized ecommerce customer references remain quarantined because inventing a customer mapping would conceal the source defect.

Source	Records read	Accepted records	Quarantined records
Customers	400	400	0
POS	1,051	1,048	3
Ecommerce orders	265	226	39

The three POS rejects were duplicates. Ecommerce rejection reasons included 35 unknown-customer references and five duplicate-order flags across 39 rejected orders; one record had overlapping reasons. All 226 accepted orders reconciled to item totals and expanded into 553 lines. Together with POS, they produced 1,601 accepted lines. These are batch-level results; duplicate detection does not establish a historical, cross-batch deduplication service.

A controlled recovery demonstration changed the first item quantity in a copy of order 900000 from its original value 1 to -1. Validation rejected the test copy. Restoring the authoritative original quantity admitted one order and four reconciled lines into separate demonstration paths. Production data was not modified, and the 42 production rejects were not repaired by this demonstration.

Curated storage and analytical findings

Athena CTAS wrote Snappy-compressed Parquet to curated/sales, partitioned by sale_year and sale_month. The verified sample resides in sale_year=2026/sale_month=08. Curated omits customer first and last names but retains customer identifiers and segmentation fields, so it is not anonymous. The table is explicitly shaped through SQL rather than relying on crawler inference alone.

Six query result sets support five business questions, with product and category analyses grouped together. Daily, regional, loyalty and category totals reconcile to 320,570.35. Store totals reconcile to POS sales only. The top-ten-product result is a ranking and must not be interpreted as total portfolio revenue.

Business question	Verified finding
How do daily store and ecommerce sales compare	POS exceeds ecommerce on each of the seven days; ecommerce contributes 33.1 percent of total sales
Which regions lead by channel	Ontario leads POS; Quebec leads ecommerce; Atlantic ecommerce exceeds Atlantic POS
Which products and categories generate the most sales	Alpine Hiking Shoe leads product revenue; Camping leads category revenue at 84,458.34
How do customer segments differ	Standard contributes the most sales; Silver has the highest average transaction value in the reporting week
Which stores lead POS sales	Toronto Yonge St leads Victoria Douglas by 37.41; Victoria has more transactions and units

QuickSight access was denied for the Learner Lab identity. We used a pandas and matplotlib notebook in VS Code, with saved charts and a browser-readable HTML version. It consumes exported Athena results and does not connect live to AWS. Five original CSV exports are retained unchanged; the category CSV was transcribed from the complete Athena results supplied during verification. The category query and mapping rationale accompany the package.

These findings describe one week and do not establish growth trends or causal loyalty effects. Customer counts by tier cannot establish cross-channel customer overlap. Store productivity cannot be inferred without traffic, floor area or operating hours. The category matching provides full sample coverage but still needs an authoritative product master for a larger deployment.

Governance and operational controls

Bucket Block Public Access is enabled, and default encryption uses SSE-S3. The deployed DenyInsecureTransport policy rejects insecure transport for the bucket and objects, with direct AWS service principals excluded from that deny. CloudTrail recorded a successful PutBucketPolicy request with HTTP 204. Default encryption was inspected at bucket level, not audited object by object.

The lab uses existing execution roles and restricts IAM administration. An iam:GetPolicy request was explicitly denied, and Lake Formation ListPermissions was unavailable. We documented those boundaries rather than claiming to have inspected effective permissions. A failed permissions listing does not prove that no grants exist.

The governance package contains a proposed analyst IAM policy, additional bucket restrictions, a Lake Formation permission set and a RACI. The hypothetical ThornquistAnalyst receives Curated SELECT and metadata access through a dedicated Athena workgroup, with restricted query-result storage. The proposed Lake Formation design requires registered Curated storage and an appropriate registration role. These analyst controls were not deployed. The live bucket policy proves transport protection, not demonstrated analyst cross-zone isolation. Administrator review and positive and negative access tests remain prerequisites.

Operational monitoring includes Glue logs and metrics, an enabled Lambda Errors alarm, and an enabled EventBridge rule for FAILED or TIMEOUT states from both Glue jobs. The rule targets thornquist-pipeline-alerts. Its SNS policy permits EventBridge publishing without creating another execution role. The observed service permission is not restricted exclusively to this rule.

Seven AWS pattern tests matched both failure states for each job and rejected successful runs and an unrelated job. A direct SNS test email was received. These separately verify filtering and SNS delivery; no real Glue failure-to-email or Lambda alarm-to-email transition was tested. Final Lambda statistic and missing-data settings also require direct verification. CloudTrail Event history supplies management audit evidence, without proving a dedicated trail or S3 object-access logging.

Cost performance and remaining work

Serverless storage and on-demand processing fit the short lab sessions and small sample. S3 Standard supports immediate interactive access; no lifecycle transition was implemented because this project has not established a retention period or an older-partition access pattern. Monthly partitions avoid excessive fragmentation at this scale. Future queries should explicitly filter partition columns when selecting periods. Parquet can reduce column reads, and partition filters can exclude other periods as data accumulates [2,3].

The current sample occupies only one monthly partition, so partitioning has not demonstrated a measured multi-period scan reduction. No CSV-versus-Parquet benchmark or dollar saving is claimed. Both Glue jobs use

version 4.0 with two G.1X workers. The successful sales run reported 0.062222 DPU-hours and the customer run 0.046111. Those observations are not a complete project bill. Repeated crawls, retained objects and query executions must also be considered. An exact cost estimate should use recorded usage and the applicable regional prices [4].

The main lessons were to inspect inferred schemas before trusting catalog types, reconcile at the correct order and line grain, and distinguish configuration evidence from execution evidence. Customer Parquet ingestion required adapting the sales reader. Ecommerce category coverage required an explicit product-name derivation rather than silently excluding online sales. The notebook fallback retained analytical coverage when QuickSight was unavailable.

Before submission, the team must assemble current scripts and SQL in the repository, reconcile screenshot filenames, record each stage executing, and rehearse the eight-minute demonstration. A clean-session reproduction has not yet been demonstrated. Further engineering should automate NDJSON and Curated refresh where needed, replace independent timers with success-dependent orchestration, verify scheduled runs and protected POS schema, and test approved analyst access and complete alert delivery. Alahi leads foundation and documentation coordination, Justin contributes ingestion and quality logic, and Seyi leads analytics, scheduling and integration verification. Shared review should preserve those contributions in the final repository.

References

[1] Humber College. CCGC 5506 Mid Term Project. Project 1 Thornquist Outfitters. Fall 2026. Requirements, assessment rubric and Learner Lab constraints.

[2] AWS. Partition your data. https://docs.aws.amazon.com/athena/latest/ug/partitions.html

[3] AWS. Optimize data. https://docs.aws.amazon.com/athena/latest/ug/performance-tuning-data-optimization-techniques.html

[4] AWS. AWS Glue pricing. https://aws.amazon.com/glue/pricing/

[5] AWS. Automating AWS Glue with EventBridge. https://docs.aws.amazon.com/glue/latest/dg/automating-awsglue-with-cloudwatch-events.html

[6] Project evidence. Verified AWS screenshots and query results, sales_model_summary.json, category mapping results, and seven EventBridge pattern test results, 22 October 2026. Evidence files are indexed separately in the governance package.